

FEATURE COMPRESSION AND OPTIMIZATION

Objective

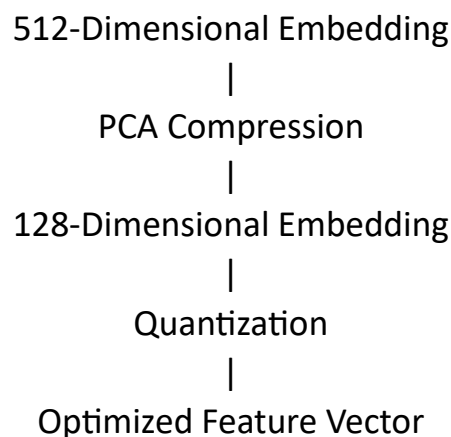
The objective of this work is to reduce the size of iris feature vectors before applying homomorphic encryption. Feature compression helps reduce storage requirements and computational cost while preserving important information required for iris authentication.

PCA Based Feature Compression

In this notebook, Principal Component Analysis (PCA) was used to compress the iris feature embeddings generated using ResNet50 in the previous phase. The generated embeddings contained 512 dimensions. PCA was applied to reduce the dimensionality while preserving most of the important information.

The compressed embeddings can be used in the next phase for homomorphic encryption.

Workflow:



PCA Compression Results

- 1) Successfully loaded the ResNet50 feature embeddings.
- 2) Applied PCA on the generated iris embeddings.
- 3) Reduced the feature dimensions from 512 to 128.
- 4) Achieved 91.67% variance retention after compression.
- 5) Generated PCA variance retention graph.
- 6) Saved compressed embeddings for future processing.

Quantization

After PCA compression, quantization was applied to further optimize the feature vectors. The compressed embeddings were converted from float32 format to int16 format. This helps reduce memory requirements.

Quantization Results

- 1) Successfully converted compressed embeddings into int16 format.
- 2) Reduced memory usage of the feature vectors.
- 3) Generated optimized feature vectors for encryption.
- 4) Saved quantized embeddings for the next phase.

Compression Statistics

- 1) Original Dimensions: 512 ->> Compressed Dimensions: 128
- 2) Variance Retained: 91.67%
- 3) Dimension Reduction: 75%

Observations

PCA significantly reduced the feature dimensions while preserving most of the important information. The compression process reduced storage requirements and computational cost. Quantization further optimized the embeddings and prepared them for homomorphic encryption.

Conclusion

- > Successfully implemented PCA-based feature compression.
- > Reduced iris embedding dimensions from 512 to 128.
- > Achieved 91.67% variance retention after compression.
- > Applied quantization to optimize the feature vectors.
- > Generated compressed embeddings for homomorphic encryption.